Teaching to the Test, Undetected: No Release-Localized Contamination Premium on Public versus Held-Out Benchmarks

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analysis pass of record: 2026-07, seed 0 (wild-cluster bootstrap only; randomization inference is exhaustive)

Abstract

In Epoch AI's cross-benchmark capability panel, models released after a benchmark's publication score above their general-capability (ECI) expectation on *public* benchmarks relative to four *held-out* benchmarks — but the premium does not survive cluster-honest inference, and the design's own placebo battery shows it is not localized at benchmark release. In 896 model × benchmark cells spanning the 19 benchmarks that straddle their own release date, the post × public level contrast is $\tau = +0.127$ z (CR1 SE 0.079); every level specification is positive, and late models underperform their ECI expectation on held-out benchmarks by -0.19 z — exactly the sign pattern a teaching-to-the-test story predicts. But with only 4 held-out benchmark clusters the inference of record is a wild-cluster bootstrap ($p = 0.145$) and exact randomization inference over all $\binom{19}{4} = 3{,}876$ held-out labelings (one-sided $p = 0.213$), neither of which rejects. The premium collapses under IRT-difficulty controls (+0.04) and model fixed effects (+0.03) — held-out status is confounded with benchmark difficulty — and it fails the placebo-cutoff localization test: the contrast at a -0.5-year placebo cutoff (+0.267) *exceeds* the true-cutoff estimate, the signature of a smooth group trend difference in event time rather than a step at release. Ability-proxy variants that avoid ECI's mechanical attenuation (the index is trained on the public benchmarks) triple the estimate (+0.35 to +0.39, wild-cluster $p = 0.047$–0.082) but inherit the same localization failure. A difference-in-kinks secondary is a correctly identified artifact (its placebo cutoff beats every true-cutoff estimate), and the automated `natex` SuDDDS run recovered the declared contrast but honestly *refused* a placebo p (no in-space placebos with a single binary group dimension); the manual bootstrap/randomization battery fills exactly that gap and does not reject. The verdict is a suggestive-sign null: no release-localized contamination premium is detectable in this panel.

1 Introduction

Train–test contamination — benchmark answers leaking into the training corpora of the models being benchmarked — is the standing objection to public AI leaderboards. [9] show that test-set membership can be proven from a model's preference for the canonical ordering of an exchangeable benchmark; [11] exploit the natural experiment of GPT training cutoffs to show that performance on code benchmarks is systematically related to whether the problems predate the cutoff; [13] commission a fresh clone of GSM8k and find accuracy drops of up to 13% for model families that apparently overfit the public original. [4] generalize the worry beyond verbatim leakage: *training*

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on the test task — benchmark-shaped data and formats entering training pipelines — confounds model comparisons even when no test item was ever seen.

The benchmark-building community's institutional response has been to hold answers back. FrontierMath keeps private problem tiers scored only via Epoch AI [6]; Humanity's Last Exam withholds a private split against overfitting [10]; ARC-AGI-2 maintains a private evaluation set [3]; LiveBench rotates fresh questions monthly precisely to stay ahead of training corpora [12]. This design pattern implies a cross-benchmark test of contamination in the wild. If public-benchmark scores are inflated by leakage or test-task training, then models released *after* a public benchmark's publication should overperform their general capability on it, and the premium should be absent on held-out benchmarks whose answers cannot be downloaded — a difference-in-differences in event time around each benchmark's release date, with held-out benchmarks as the comparison group.

This paper runs that test on Epoch AI's benchmarking panel [5, 8], treating benchmark release as a dated event inside a model × benchmark panel and demanding that any premium be *localized* at the event: estimated with cluster-honest inference (only 4 of 19 benchmark clusters are held out), stress-tested against difficulty controls, and benchmarked against placebo release dates, with the automated `natex` toolkit's discovery-and-refusal machinery [7] reported as run. The sign pattern survives everything; the localized premium survives nothing.

2 Data

Three public Epoch AI files are joined [5]: the ECI benchmark score table (2,010 model × benchmark performance cells with benchmark release dates), the ECI model index (211 models with release dates and ECI capability scores [8]), and the IRT benchmark parameter table (52 benchmarks with estimated difficulty `edi` and discrimination). Three release dates missing from the source are hand-filled as declared inputs: HLE 2025-01-23, ARC-AGI-2 2025-03-24, and FrontierMath-Tier-4-Private 2025-07-01. The held-out classification is likewise declared, by whether answers are publicly downloadable: FrontierMath-2025-02-28-Private, FrontierMath-Tier-4-2025-07-01-Private, HLE, and ARC-AGI-2 (4 benchmarks); all other benchmarks are public.

A cell is a model × benchmark pair. The running variable is event time, x = model release date − benchmark release date, in years. The identifying sample keeps the 19 benchmarks with cells on both sides of $x = 0$: $n = 896$ cells across 150 models, split 227 pre-release / 668 post-release (one cell at exactly $x = 0$ is coded pre). Outcomes are built within benchmark: z is the within-benchmark z-score of performance, and the primary outcome r residualizes z linearly on the model's ECI score, so r measures performance relative to what the model's general capability predicts on that benchmark. Variants used below: a logit-performance residual (48 zero and 3 one scores clipped at $[0.01, 0.99]$), and two ability proxies that avoid ECI itself — the model's mean z across all *other* benchmarks (leave-one-out) and across held-out benchmarks only. The panel builder asserts bit-identical reproduction of the scout-pass r before writing any file.

3 Design and Methods

Level contrast. The primary estimand is the post-release premium of public over held-out benchmarks in r, from the cell-level regression

$$r_{mb} = \alpha + \beta x_{mb} + \gamma\,\mathrm{post}_{mb} + \tau\,(\mathrm{post}_{mb} \times \mathrm{public}_b) + \mu_b + u_{mb},$$

with benchmark fixed effects μ_b and a common linear event-time trend; γ is the held-out post-release baseline and τ the contamination premium. Errors are CR1 cluster-robust by benchmark with a

$t(18)$ reference. With 19 clusters of which only 4 are held out, CR1 alone is not trustworthy [2], so the inference of record is (i) a null-imposed Rademacher wild-cluster bootstrap (B = 9,999, seed 0) and (ii) *exact* randomization inference over all $\binom{19}{4} = 3{,}876$ assignments of the held-out label to 4 of the 19 benchmarks, FWL-accelerated (the observed labeling's coefficient is ranked in the full permutation distribution; no sampling anywhere).

Localization. A contamination premium must be a step at benchmark release, not a generic divergence of the two groups in event time. The identical regression is re-estimated with the cutoff moved to $x_0 \in \{-1.0, -0.5, +0.5, +1.0\}$ years; a true release-localized effect should peak at the true cutoff. Side composition (straddling benchmarks and held-out benchmarks per placebo) is reported because extreme placebo cutoffs thin one side.

Robustness. Seven re-estimates: the logit-residual outcome; post $\times$ IRT-difficulty and post $\times$ discrimination controls (standardized `edi` and slope); a two-way model + benchmark fixed-effects specification on z; the two ability-proxy outcomes; dropping saturated cells (performance > 0.95); and model-release-year fixed effects, which isolate cross-benchmark stagger from calendar time. A per-benchmark jackknife re-estimates τ dropping one cluster at a time.

Automated discovery (SuDDDS). The `natex discover` pipeline [7] ran on the benchmark $\times$ half-year-bin panel (94 cells; `--design did`, declared treatment path θ = public $\times$ post, seed 0). Its subset scan, anticipation gate, and placebo machinery either validate the declared event or refuse; refusals are reported as run, per the toolkit's honest-inference conventions.

Difference-in-kinks secondary. As a slope-based cross-check, a group difference-in-kinks in the tradition of [1] — `natex kink` with the held-out group as the comparison dimension, triangular kernel, CR1 by benchmark — estimates the change in the event-time *slope* contrast at $x = 0$, over bandwidths 0.5–2.0 years, with a placebo cutoff at $x_0 = +0.5$.

4 Results

A positive premium with a contamination-consistent sign pattern. The primary contrast is $\tau = +0.1267$ z (CR1 SE 0.0786, $t = 1.61$, $p_{t(18)} = 0.124$) on $n = 896$ cells in 19 benchmark clusters. The held-out post-release baseline is $\gamma = -0.1929$ z (SE 0.0526): after a benchmark's release, models underperform their ECI expectation on held-out benchmarks, while public benchmarks absorb the offsetting premium — the two-sided pattern teaching-to-the-test predicts. The premium is not driven by any single cluster: the per-benchmark jackknife keeps $\tau \in [+0.085, +0.171]$, positive in all 19 leave-one-out samples.

Cluster-honest inference does not reject. The wild-cluster bootstrap gives $p = 0.145$. Exact randomization inference over all 3,876 held-out labelings gives one-sided $p = 0.213$ (two-sided 0.384): more than a fifth of all possible "held-out" labelings produce a premium at least as large as the observed one, and the permutation distribution's 5th/50th/95th percentiles (-0.224 / $+0.028$ / $+0.200$) comfortably bracket $+0.127$. With 4 treated-side clusters, this is the honest finite-sample yardstick, and the observed contrast is unremarkable against it.

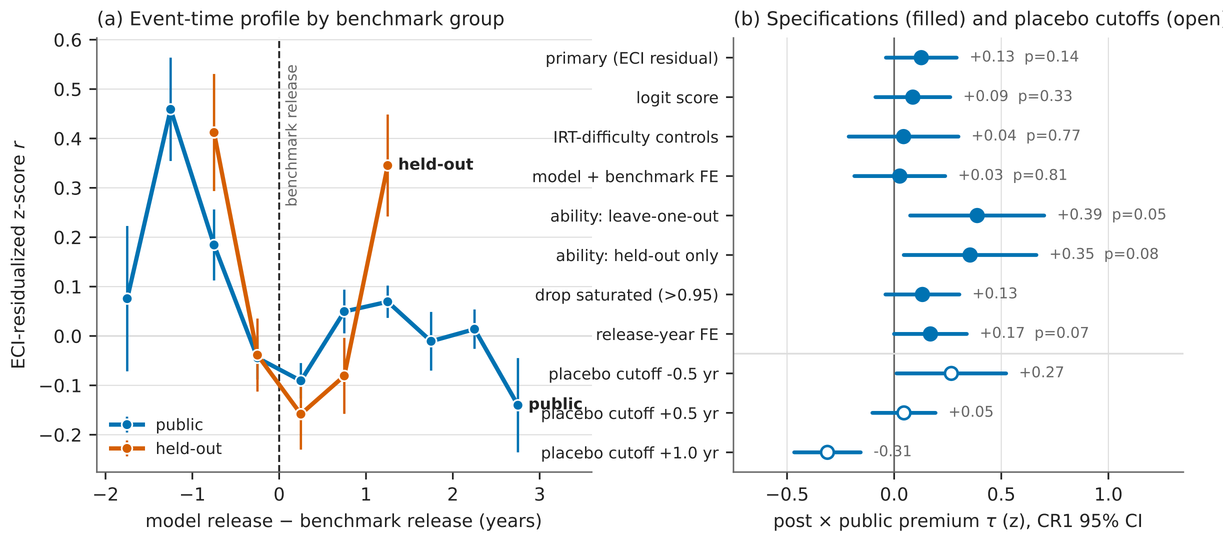

Figure 1: (a) Mean ECI-residualized within-benchmark z-score r by 0.5-year event-time bin (whiskers ± 1 SE; bins with ≥ 3 cells), public versus held-out benchmarks, vertical rule at benchmark release. In the first year after release, held-out cells fall below their capability expectation while public cells sit above it — but the raw profiles show no step at $x = 0$, and the two groups already differ before release. (b) The post $\times$ public contrast τ across specifications (filled circles; CR1 95% CIs, $t(18)$ critical value; wild-cluster bootstrap p annotated where computed) and at placebo event-time cutoffs (open circles). Every specification is positive, but the -0.5-year placebo cutoff exceeds the true-cutoff estimate — the localization failure that grades this a null.

The premium is difficulty in disguise. The held-out set — FrontierMath tiers, HLE, ARC-AGI-2 — is also the hardest-benchmark set. Adding post $\times$ IRT-difficulty and post $\times$ discrimination controls collapses τ from $+0.127$ to $+0.044$ (wild-cluster $p = 0.769$); two-way model + benchmark fixed effects give $+0.026$ ($p = 0.814$); the logit outcome gives $+0.087$ ($p = 0.334$). Two specifications move the other way: the ability-proxy outcomes, which replace ECI — itself fit *on the public benchmarks* [8] and therefore mechanically absorbing part of any public-benchmark inflation into measured capability — with leave-one-out or held-out-only mean z. These triple the estimate ($+0.388$, wild-cluster $p = 0.047$; $+0.354$, $p = 0.082$), confirming the attenuation channel is real, but they inherit the localization failure below and the 4-cluster fragility. Dropping saturated cells ($+0.132$) and release-year fixed effects ($+0.169$, $p = 0.072$) leave the picture unchanged.

Localization fails. Moving the cutoff half a year *before* benchmark release yields $\tau = +0.267$ (SE 0.122, 15 straddling benchmarks, all 4 held-out) — *larger* than the true-cutoff estimate. The grid declines monotonically through the true cutoff: $+0.800$ at -1.0 (uninterpretable: 11 pre cells, 4 straddling benchmarks), $+0.267$ at -0.5, $+0.127$ at 0, $+0.046$ at $+0.5$, -0.311 at $+1.0$. A release-localized contamination premium should peak at release; a smooth difference in the two groups' event-time trends produces exactly this monotone drift. The design therefore identifies the positive contrast as a trend difference, not a step at benchmark release.

Automated runs. The `natex` SuDDDS pipeline discovered the declared subset {group: public} at $t_0 = 0$ with window 2 (treatment-path scan $p = 0.010$ — mechanical given the declared θ) and estimated DD $\tau = +0.232$ (SE 0.088) on the 94-cell binned panel, but *refused* its placebo

Table 1: Headline estimates. Outcome r is the within-benchmark z-score of performance residualized on model ECI; τ is the post $\times$ public coefficient with benchmark FE and a linear event-time trend, CR1 clustered by benchmark (19 clusters, 4 held-out). WCB: null-imposed Rademacher wild-cluster bootstrap, $B = 9{,}999$, seed 0. RI: exact randomization inference over all $\binom{19}{4} = 3{,}876$ held-out labelings.

Quantity	τ	CR1 SE	p	notes
Panel A: primary contrast and inference of record				
post $\times$ public (primary)	+0.127	0.079	0.145	WCB; $p_{t(18)} = 0.124$
exact RI, one-sided	—	—	0.213	two-sided 0.384
held-out post baseline γ	−0.193	0.053	—	contamination-consistent sign
Panel B: robustness (WCB p where computed)				
logit score	+0.087	0.083	0.334	$n = 896$
IRT-difficulty controls	+0.044	0.122	0.769	post $\times$ edi, post $\times$ slope
model + benchmark FE	+0.026	0.101	0.814	outcome z
ability: leave-one-out	+0.388	0.149	0.047	$n = 895$
ability: held-out only	+0.354	0.147	0.082	$n = 661$
drop saturated (perf > 0.95)	+0.132	0.082	—	$n = 878$
release-year FE	+0.169	0.081	0.072	$n = 896$
Panel C: placebo event-time cutoffs (straddling / held-out)				
cutoff −1.0 yr	+0.800	0.078	—	4/1; 11 pre cells, uninterpretable
cutoff −0.5 yr	+0.267	0.122	—	15/4; *exceeds true cutoff*
cutoff +0.5 yr	+0.046	0.070	—	19/4
cutoff +1.0 yr	−0.311	0.074	—	10/3
Panel D: automated `natex` runs				
SuDDDS DD (94-cell panel)	+0.232	0.088	refused	0 in-space placebos; gate closed
DiK slope, cutoff 0, bw 0.75	+0.015	0.443	—	drifts to −0.843 (0.218) at bw 2.0
DiK placebo +0.5, bw 0.75	−1.467	0.644	—	exceeds all true-cutoff estimates

Notes: jackknife over benchmarks keeps $\tau \in [+0.085, +0.171]$ (positive in all 19 leave-one-out samples). RI permutation quantiles (5%/50%/95%): −0.224 / +0.028 / +0.200. The SuDDDS scan $p = 0.010$ localizes the declared treatment path and is mechanical given the declared θ; its refusal of a placebo p is the design working as intended with one binary group dimension.

randomization p: with a single binary group dimension there are 0 usable in-space placebos (≥ 5 required), and the anticipation gate failed closed (null Holm p-values). The manual wild-cluster-bootstrap and exact-randomization battery above is precisely the placebo inference the refusal requests, and it does not reject. The difference-in-kinks secondary is an identified artifact: the slope contrast at the true cutoff is +0.015 (SE 0.443) at bandwidth 0.75, drifting to −0.843 (SE 0.218) at bandwidth 2.0 as saturated public post-release profiles compress, while the +0.5 placebo cutoff gives −1.467 (SE 0.644) — larger in magnitude than every true-cutoff estimate — so the slope design carries no evidential weight here.

5 Caveats and Conclusion

Six caveats bound the claims, none softened. First, *four held-out clusters*: CR1/$t(18)$ inference alone is not trustworthy at this cluster count [2]; the wild-cluster bootstrap and the exact label-randomization test are the inference of record, and they do not reject — but randomization inference assumes label exchangeability, which the composition differences below strain; it remains the most honest finite-sample check available here. Second, *difficulty confounding*: held-out status is not randomly assigned — the held-out set is the hardest-benchmark set, and post $\times$ difficulty controls absorb most of the premium. Third, *mechanical attenuation*: ECI is trained on the public benchmarks, so contamination inflates the capability control itself; the ability-proxy variants quantify this (tripling τ) but inherit every other weakness. Fourth, *localization failure*: the -0.5-year placebo contrast exceeds the true-cutoff contrast, so the design cannot distinguish a release-localized step from a smooth group trend difference — and reads the data as the latter. Fifth, *time confounding*: within a benchmark, event time and calendar time are collinear; the release-year-FE specification leans entirely on cross-benchmark stagger. Sixth, *declared inputs and ceilings*: three hand-filled release dates and the held-out membership are declared, not estimated; saturation ceilings compress public post-release residuals *against* the hypothesis, and the logit and drop-saturated variants only partially derate this, so a true premium could be somewhat attenuated — but ceilings cannot manufacture the placebo-cutoff pattern.

The verdict is a suggestive-sign null. Every level specification is positive; the held-out post-release shortfall (-0.19 z) is exactly the shape leakage or test-task training [4] would leave; and the ability-proxy estimates show the premium is larger once the contaminated capability index is removed from the control set. But the premium is not significant under any cluster-honest test, collapses when held-out difficulty is controlled, and — decisively — is not localized at benchmark release. What these data support is at most a slow divergence of public-benchmark performance from held-out performance across model generations, consistent with gradual test-task adaptation [4] and with [13]'s finding that frontier models show little sharp overfitting; what they do not support is a detectable step premium that switches on when a benchmark's answers become available. Benchmark designers' held-out and rotating designs [6, 10, 3, 12] are the reason this test was possible at all; more held-out clusters would make it decisive.

Reproducibility. All estimates derive from the public Epoch AI tables plus three declared release dates; no dataset files are committed. The only stochastic step is the wild-cluster bootstrap (single generator, seed 0); randomization inference is exhaustive and `natex` runs declared seed 0. Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates, the placebo grid, the jackknife range, and the panel composition against the numbers of record before drawing.

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